Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Full-Stack · TypeScript · React · AI-Augmented Engineering · Testing Strategy · CI/CD · Scalable Systems · Python · FastAPI · AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years building production web applications across full-stack environments - with a strong focus on technical quality, testing strategy, and AI-augmented development. At Entrata, owned architecture and delivery of 20+ scalable production modules across 120+ Agile sprints, enforced CI/CD-driven quality standards (sub-10 bugs for 6+ consecutive months), and led 300+ code reviews raising the engineering bar across the team. Built AgentProbe - an AI-augmented testing and evaluation platform - demonstrating the exact pattern this role values: using LLMs as force multipliers while maintaining high standards for correctness, maintainability, and production hardening. Full-stack experience spans Python/FastAPI backends, TypeScript/JavaScript, and React (AI on FHIR). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Full-Stack & Frontend: TypeScript, JavaScript, React, HTML, CSS, FastAPI, PHP, Python, Bash, Docker, Microservices, Scalable Systems, REST APIs, CI/CD (GitHub Actions), Unit Testing, Integration Testing, End-to-End Testing, Performance Optimization, AWS (S3, EC2, RDS), Redis, PostgreSQL, MySQL, Oracle, Linux, Git

AI-Augmented Engineering: AI-Assisted Development, LLM-as-Judge, LangGraph, LangChain, Agentic AI, RAG Pipelines, ChromaDB, Model Evaluation, Hallucination Detection, HITL, Prompt Engineering, OpenAI API, Anthropic API, MCP

Quality & Architecture: Testing Strategy (unit, integration, UI, end-to-end), CI-Driven Development, Code Review, Design Reviews, Production Readiness, Reliability Engineering, Observability, Agile/Scrum, Technical Mentorship

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built AI-augmented full-stack backend using LangGraph and Python - prototyped rapidly with AI tools, refined to production-hardened system with validated outputs; testing strategy enforcing correctness before every deployment.
- Implemented RAG pipeline with end-to-end quality controls; AI-assisted development with strong judgment on when to rely on LLM output vs. deterministic logic.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built scalable full-stack data pipelines (Python, R, TypeScript) with end-to-end testing and statistical validation - $R^2 = 0.977$ across 190+ countries; mentored 600+ students in production-quality engineering practices.
- Designed monitoring dashboards improving research comprehension by 40%; led technical sessions on software design and testing methodology.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned technical quality across architecture, design reviews, and CI/CD** - 300+ code reviews enforcing maintainability and correctness standards; sub-10 bug count for 6+ consecutive months; 99.5% uptime SLA at 100K+ monthly active users across 120+ Agile sprints.
- **Architected scalable, modular full-stack systems** - 20+ production modules spanning Python/FastAPI backends, JavaScript/TypeScript, and PostgreSQL/MySQL data layers; Docker microservices redesign (50% scalability improvement, 30% deployment reduction); event-driven architecture with RabbitMQ and Redis.
- **Built and enforced robust testing strategy** - unit, integration, and functional test suites across compliance-critical workflows; CI-driven development with automated testing pipelines eliminating manual verification; observability tooling reducing triage time 40%.
- **Performance and reliability improvements** - 200+ SQL query optimizations (30% gain); AWS S3, EC2, RDS at production scale; 100% PHI compliance (zero security incidents); scalable data pipelines processing 180,000 domain records per release cycle.
- Mentored 5+ engineers through design reviews and technical guidance; drove cross-team alignment on engineering standards and best practices.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built full-stack property management modules (PHP, Python, JavaScript, PostgreSQL); scalable data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI-Augmented Testing & Quality Platform

AI-Augmented · Testing

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

AI-augmented engineering platform: prototype → refinement → production hardening using LLMs as force multipliers. 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, LLM-as-judge auto-rater with strong correctness standards. 1,240+ test cases - zero regressions shipped to production.

AI on FHIR: Full-Stack React + Python Healthcare Platform

Full-Stack · React

Python · Flask · TypeScript · React · LLMs · FHIR · Docker · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack TypeScript/React + Python platform translating natural-language queries into FHIR API calls via LLM inference and BERT embeddings - reduced retrieval time ~60%; modular, scalable architecture with HIPAA-compliant RBAC and full data isolation.

Leasel: Scalable Agentic Backend with Production Hardening

Backend · Reliability

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI-augmented backend: rapid prototyping with LLMs converging on production-hardened system with deterministic reliability controls, HITL escalation, and CI quality gate. 93.6% precision, 87.9% F1 across 1,240 test cases; full observability and telemetry.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017